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Semiconductor device structure

Abstract

Semiconductor structures and the manufacturing method thereof are disclosed. An exemplary semiconductor structure according to the present disclosure includes a substrate having a p-type well or an n-type well, a first base portion over the p-type well, a second base portion over the n-type well, a first plurality of channel members over the first base portion, a second plurality of channel members over the second base portion, an isolation feature disposed between the first base portion and the second base portion, and a deep isolation structure in the substrate disposed below the isolation feature.

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Background/Summary

PRIORITY DATA (1) This application is a continuation application of U.S. patent application Ser. No. 17/871,697, filed Jul. 22, 2022, which is a divisional application of U.S. patent application Ser. No. 17/142,970, filed Jan. 6, 2021, which claims priority to U.S. Provisional Patent Application No. 63/106,724, filed on Oct. 28, 2020, entitled “Semiconductor Device Structure”, each of which is hereby incorporated herein by reference in its entirety.

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

(2) For example, as integrated circuit (IC) technologies progress towards smaller technology nodes, multi-gate metal-oxide-semiconductor field effect transistor (multi-gate MOSFET, or multi-gate devices) have been introduced to improve gate control by increasing gate-channel coupling, reducing off-state current, and reducing short-channel effects (SCEs). A multi-gate device generally refers to a device having a gate structure, or portion thereof, disposed over more than one side of a channel region. Fin-like field effect transistors (FinFETs) and multi-bridge-channel (MBC) transistors are examples of multi-gate devices that have become popular and promising candidates for high performance and low leakage applications. A FinFET has an elevated channel wrapped by a gate on more than one side (for example, the gate wraps a top and sidewalls of a “fin” of semiconductor material extending from a substrate). An MBC transistor has a gate structure that can extend, partially or fully, around a channel region to provide access to the channel region on two or more sides. Because its gate structure surrounds the channel regions, an MBC transistor may also be referred to as a surrounding gate transistor (SGT) or a gate-all-around (GAA) transistor.

(3) While MBC transistors with sheet-like channel members generally provide superior gate control and drive current, their wider sheet-like channel members may increase device widths. Such increased device widths may make them less attractive in high packing density applications, such

as memory applications. Measures to improve packing density of MBC transistors may face challenges in forming isolation structures to isolate different device regions. While existing semiconductor devices are generally adequate for their intended purposes, they are not satisfactory in all aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a flowchart of a method for forming a semiconductor device, according to one or more aspects of the present disclosure.

(3) FIGS. 2-23 illustrate fragmentary perspective views or cross-sectional views of a workpiece during various fabrication stages in the method of FIG. 1, according to one or more aspects of the present disclosure.

DETAILED DESCRIPTION

(4) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(5) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(6) Further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range considering variations that inherently arise during manufacturing as understood by one of ordinary skill in the art. For example, the number or range of numbers encompasses a reasonable range including the number described, such as within $\pm 10\%$ of the number described, based on known manufacturing tolerances associated with manufacturing a feature having a characteristic associated with the number. For example, a material layer having a thickness of “about 5 nm” can encompass a dimension range from 4.25 nm to 5.75 nm where manufacturing tolerances associated with depositing the material layer are known to be $\pm 1-15\%$ by one of ordinary skill in the art. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(7) The present disclosure is generally related to isolation structures to reduce bulk leakage, and more particularly to deep isolation structures disposed on well junctions.

(8) To improve drive current to meet design needs, MBC transistors may include nanoscale channel members that are thin and wide. Such MBC transistors may also be referred to as nanosheet transistors. While nanosheet transistors are able to provide satisfactory drive current and channel control, their wider nanosheet channel members may make it challenging to further reduce cell size. In some example structures, fish-bone structures or fork-sheet structures may be implemented to reduce cell dimensions. In a fish-bone structure or a fork-sheet structure, adjacent stacks of channel members may be divided by dielectric fins (or hybrid fins). As one end of the stacks of channel members are in contact with the dielectric fins, a gate structure that wraps around the stacks of channel members does not extend between the channel members and the dielectric fin. Each of the dielectric fins is disposed on an isolation feature, such as a shallow trench isolation (STI) feature. An STI feature may also be disposed on a junction between an n-type well and a p-type well to reduce bulk leakage.

(9) The present disclosure provides a deep isolation structure that is disposed at a well junction between a p-type well under an n-type device and an n-type well under a p-type device. In some embodiments, the formation of the deep isolation structure includes formation of a notch along the well junction. The notch may undercut active regions of the n-type device and the p-type device. In at least some embodiments, the n-type device and p-type device may be fish-bone or fork-sheet transistors. The deep isolation structure may better block the bulk leakage path between the n-type well and the p-type well.

(10) The various aspects of the present disclosure will now be described in more detail with reference to the figures. FIG. 1 illustrates a flowchart of a method **100** of forming a semiconductor device. Method **100** is merely an example and is not intended to limit the present disclosure to what is explicitly illustrated in method **100**. Additional steps may be provided before, during and after method **100**, and some steps described can be replaced, eliminated, or moved around for additional embodiments of the methods. Not all steps are described herein in detail for reasons of simplicity. Method **100** is described below in conjunction with FIGS. 2-23, which illustrate fragmentary cross-sectional views of a workpiece **200** at different stages of fabrication according to embodiments of method **100**. Because a semiconductor device will be formed from the workpiece **200**, the workpiece **200** may be referred to as a semiconductor device **200** as the context requires. Although embodiments that include fish-bone or fork-sheet transistors are illustrated in the figures, the present disclosure is not so limited and may be applicable to other multi-gate devices, such as MBC transistors or FinFETs. Throughout FIGS. 2-23, the X direction, the Y direction, and the Z direction are perpendicular to one another and are used consistently. Additionally, throughout the present disclosure, like reference numerals are used to denote like features.

(11) Referring to FIGS. 1 and 2, method **100** includes a block **102** where a workpiece **200** is received. As shown in FIG. 2, the workpiece **200** includes a substrate **202** and a stack **204** disposed on the substrate **202**. In one embodiment, the substrate **202** may be a silicon (Si) substrate. In some other embodiments, the substrate **202** may include other semiconductor materials such as germanium (Ge), silicon germanium (SiGe), or a III-V semiconductor material. Example III-V semiconductor materials may include gallium arsenide (GaAs), indium phosphide (InP), gallium phosphide (GaP), gallium nitride (GaN), gallium arsenide phosphide (GaAsP), aluminum indium arsenide (AlInAs), aluminum gallium arsenide (AlGaAs), gallium indium phosphide (GaInP), and indium gallium arsenide (InGaAs). The substrate **202** may include multiple n-type well regions and multiple p-type well regions. In the depicted embodiment, the substrate **202** includes a p-type well region **202P** (or p-well **202P**) and an n-type well region **202N** (or n-well **202N**). As shown in FIG. 2, the p-well **202P** and the n-well **202N** interface one another along a well junction **2020**. The p-well **202P** may be doped with a p-type dopant (i.e., boron (B) and the n-well **202N** may be doped with an n-type dopant (i.e., phosphorus (P) or arsenic (As)). The p-well **202P** and n-well **202N** may

be formed using ion implantation or thermal diffusion.

(12) Referring still to FIG. 2, the stack **204** may include a plurality of channel layers **208** interleaved by a plurality of sacrificial layers **206**. The channel layers **208** and the sacrificial layers **206** may have different semiconductor compositions. In some implementations, the channel layers **208** are formed of silicon (Si) and sacrificial layers **206** are formed of silicon germanium (SiGe). In these implementations, the additional germanium content in the sacrificial layers **206** allow selective removal or recess of the sacrificial layers **206** without substantial damages to the channel layers **208**. In some embodiments represented in FIG. 2, the workpiece **200** also includes a top sacrificial layer **206T** disposed on the stack **204**. The top sacrificial layer **206T** is thicker than the other sacrificial layers **206** and functions to protect the stack **204** from damages during fabrication processes. The top sacrificial layer **206T**, the sacrificial layers **206** and the channel layers **208** may be deposited using an epitaxial process. The stack **204** may be epitaxially deposited using CVD deposition techniques (e.g., vapor-phase epitaxy (VPE) and/or ultra-high vacuum CVD (UHV-CVD)), molecular beam epitaxy (MBE), and/or other suitable processes. The sacrificial layers **206** and the channel layers **208** are deposited alternately, one-after-another, to form the stack **204**. FIG. 2 illustrates that three (3) layers of the sacrificial layers **206** and three (3) layers of the channel layers **208** are alternately and vertically arranged, which is for illustrative purposes only and not intended to be limiting beyond what is specifically recited in the claims. The number of layers depends on the desired number of channels members for the semiconductor device **200**. In some embodiments, the number of the channel layers **208** is between 1 and 6.

(13) Referring to FIGS. 1 and 3, method **100** includes a block **104** where the stack **204** and the substrate **202** are patterned to form fin-shaped structures **211** separated by trenches **212** or a junction trench **212J**. To pattern the stack **204** and the substrate **202**, a first hard mask layer **210** is deposited over the top sacrificial layer **206T**. The first hard mask layer **210** is then patterned to serve as an etch mask to pattern the top sacrificial layer **206**, the stack **204** and a portion of the substrate **202**. In some embodiments, the first hard mask layer **210** may be deposited using CVD, plasma-enhanced CVD (PECVD, atomic layer deposition (ALD), plasma-enhanced ALD (PEALD), or a suitable deposition method. The first hard mask layer **210** may be a single layer or a multilayer. When the first hard mask layer **210** is a multi-layer, the first hard mask layer **210** may include a pad oxide and a pad nitride layer. In an alternative embodiment, the first hard mask layer **210** may include silicon (Si). The fin-shaped structures **211** may be patterned using suitable processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a material layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned material layer using a self-aligned process. The material layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the first hard mask layer **210** and then the patterned first hard mask layer **210** may be used as an etch mask to etch the stack **204** and the substrate **202** to form fin-shaped structures **211**. The etching process can include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes.

(14) As shown in FIG. 3, each of the fin-shaped structures **211** includes a base portion **211B** formed from a portion of the substrate **202** and a top portion **211T** formed from the stack **204**. The top portion **211T** is disposed over the base portion **211B**. The fin-shaped structures **211** extend lengthwise along the Y direction and extend vertically along the Z direction from the substrate **202**. Along the X direction, the fin-shaped structures **211** are separated by trenches **212** and junction trenches **212J**. Compared to the trenches **212**, the junction trench **212J** is disposed on and along the well junction **2020**. In some embodiments represented in the FIG. 3, the junction trench **212J** is wider than a trench **212** along the X direction to provide greater spacing. As shown in FIG. 3, the junction trench **212J** defines a first spacing **S1** and the trench **212** defines a second spacing **S2**. The

first spacing S1 is greater than the second spacing S2. In some instances, the first spacing S1 is between about 20 nm and about 30 nm and the second spacing S2 is between about 10 nm and about 20 nm. The wider junction trench **212J** allows active regions, such as the fin-shaped structures **211**, on either side of the well junction **2020** to be spaced further apart to reduce bulk leakage.

(15) Referring to FIGS. **1** and **4**, method **100** includes a block **106** where a liner **214** is formed along sidewalls of the fin-shaped structure **211**. The liner **214** functions to protect sidewalls of the fin-shape structures **211** from being damaged during block **108** (to be described below). In an example process to form the liner **214**, a dielectric material is conformally deposited over the workpiece **200**, including over the trenches **212** and the junction trench **212J**, by atomic layer deposition (ALD), chemical vapor deposition (CVD), or low-pressure CVD (LPCVD). The dielectric material for the liner **214** may include silicon oxide, silicon nitride, silicon oxynitride, silicon oxycarbide, silicon carbonitride, silicon oxycarbonitride, or a combination thereof. The deposited dielectric material is then anisotropically etched back to form the liner **214**. In some embodiments, the etch back may include a dry etch process that uses one or more fluorine-containing gas, such as carbon tetrafluoride (CF₄), sulfur hexafluoride (SF₆), or nitrogen trifluoride (NF₃). As illustrated in FIG. **4**, because first spacing S1 in the junction trench **212J** is greater, the dielectric material on the bottom surface of the junction trench **212J** is removed to expose the substrate **202**. That is, the liner **214** only lines sidewalls of the junction trench **212J**. The smaller second spacing S2 of the trench **212** prevents removal of the dielectric material from its bottom surfaces.

(16) Referring to FIGS. **1** and **5**, method **100** includes a block **108** where the junction trench **212J** is extended into the substrate **202** to form a deep pocket **216**. With the trenches **212** protected by the liner **214** and the substrate **202** exposed from the bottom surface of the junction trench **212J**, operations at block **108** etches the exposed substrate **202** down the junction trench **212J**. The etching of the exposed substrate **202** extends the junction trench **212J** downward into the substrate **202** to form the deep pocket **216**. The deep pocket **216** may also be referred to as a notch. The etch at block **108** may be performed using a dry etch process, which tends to be anisotropic, or a wet etch process, which tends to be isotropic. An example selective wet etch process may include use of ethylenediamine pyrocatechol (EDP), tetramethylammonium hydroxide (TMAH), nitric acid (HNO₃), hydrofluoric acid (HF), ammonia (NH₃), ammonium fluoride (NH₄F) or a suitable wet etchant. An example selective dry etch process may include use of sulfur hexafluoride (SF₆), hydrogen (H₂), ammonia (NH₃), hydrogen fluoride (HF), carbon tetrafluoride (CF₄), argon (Ar), or a mixture thereof. In some embodiments represented in FIG. **5**, the etch process at block **108** is not perfectly anisotropic and the deep pocket **216** undercuts the liner **214**. As a result, a shape of the deep pocket **216** is different from a shape of the junction trench **212J**. In some instances, a width of the widest portion of the deep pocket **216** is greater than a width of the junction trench **212J** (the liner **214** included). The deep pocket **216** is disposed in a level below the bottom surfaces of the trenches **212** or the junction trench **212J** before the formation of the deep pocket **216**. Accordingly, the deep pocket **216** is disposed deeper in the substrate **202**.

(17) Referring to FIGS. **1**, **6** and **7**, method **100** includes a block **110** where isolation features are formed into the deep pocket **216** and trenches **212**. Isolation features formed at block **110** may include isolation features **218** in trenches **212**, and junction isolation features **218J** in the junction trench **212J**, and a deep isolation feature **220** in the deep pocket **216**. The isolation features **218** and the junction isolation feature **218J** may be collectively referred to as a shallow trench isolation (STI) feature. In an example process to form these isolation features, a dielectric material is deposited over the workpiece **200**, filling the trenches **212**, the deep pocket **216**, and the junction trenches **212J** with the dielectric material. In some embodiments, the dielectric material may include silicon oxide, silicon nitride, silicon oxynitride, silicon oxycarbide, silicon carbonitride, silicon oxycarbonitride, or a combination thereof. In various examples, at block **110**, the dielectric

material may be deposited by a CVD process, a subatmospheric CVD (SACVD) process, a flowable CVD (FCVD) process, an ALD process, spin-on coating, and/or other suitable process. The deposited dielectric material is then thinned and planarized, for example by a chemical mechanical polishing (CMP) process, until the top sacrificial layer **206T** is exposed, as shown in FIG. **6**. In some embodiments, compositions of the liner **214** and the dielectric material for the isolation features may be similar and their boundaries are marked by dotted lines. For ease of illustration, the boundaries between the liner **214** and the isolation features may be omitted in subsequent figures. Referring to FIG. **7**, the planarized dielectric material and the liner **214** is further recessed by a dry etching process, a wet etching process, and/or a combination thereof to form the final structures of the isolation features **218**, the junction isolation features **218J**, and the deep isolation feature **220**. As shown in FIG. **7**, the top portions **211T** of the fin-shaped structures **211** rise above the isolation features **218** or the junction isolation feature **218J** while the base portions **211B**, or a substantial portion thereof, is surrounded by the isolation features **218** or the junction isolation feature **218J**. As shown in FIG. **7**, the deep isolation feature **220** is disposed below the level of the base portions **211B** and may undercut base portions **211B** adjacent the well junction **2020**. That is, a portion of the deep isolation feature may extend under the adjacent base portions **211B**. After the formation of the isolation features **218** and the junction isolation feature **218J**, the top portions **211T** are separated by dielectric fin trenches **222** and a junction dielectric fin trench **222J**. The junction dielectric fin trench **222J** is disposed directly over the well junction **2020**.

(18) Referring to FIGS. **1**, **8**, **9**, **10**, and **11**, method **100** includes a block **112** where the dielectric fins are formed. In the embodiment depicted in FIG. **11**, at block **112**, dielectric fins **230** are formed in the dielectric fin trenches **222** and a junction dielectric fin **230J** is formed in the junction dielectric fin trench **222J**. An example process to form the dielectric fins are illustrated in FIGS. **8**, **9**, **10**, and **11**. Referring to FIG. **8**, a first layer **224** and a second layer **226** are conformally deposited over the workpiece **200**, including in the dielectric fin trenches **222** and the junction dielectric fin trench **222J**. The first layer **224** may be conformally deposited using CVD, ALD, or a suitable method. The first layer **224** lines the sidewalls and the bottom surfaces of the dielectric fin trenches **222** and the junction dielectric fin trench **222J**. The second layer **226** is then conformally deposited over the first layer **224** using CVD, high density plasma CVD (HDPCVD), and/or other suitable process. In some instances, a dielectric constant of the second layer **226** is smaller than that of the first layer **224**. The first layer **224** may include silicon, silicon nitride, silicon carbide, silicon carbonitride, silicon oxycarbonitride, aluminum oxide, aluminum nitride, aluminum oxynitride, zirconium oxide, zirconium nitride, zirconium aluminum oxide, hafnium oxide, or a suitable dielectric material. In one embodiment, the first layer **224** includes silicon carbonitride. The second layer **226** may include silicon oxide, silicon carbide, silicon oxynitride, silicon oxycarbonitride, or a suitable dielectric material. In one embodiment, the second layer **226** includes silicon oxide. In some embodiments represented in FIG. **8**, due to width differences, the second layer **226** completely fills the dielectric fin trenches **222** but does not completely fill the junction dielectric fin trench **222J**.

(19) Referring to FIG. **9**, the conformally deposited first layer **224** and second layer **226** are etched back to expose the top sacrificial layer **206T** and remove the first layer **224** and the second layer **226** in the junction dielectric fin trench **222J**. In some embodiments, the first layer **224** and the second layer **226** may be etched back in a dry etch process that uses oxygen, nitrogen, a fluorine-containing gas (e.g., CF_4 , SF_6 , CHF_3 , CH_2F_2 , and/or C_2F_6), a chlorine-containing gas (e.g., Cl_2 , CHCl_3 , CCl_4 , and/or BCl_3), a bromine-containing gas (e.g., HBr and/or CHBr_3), an iodine-containing gas, other suitable gases and/or plasmas, and/or combinations thereof. In some implementations, the etch back may include a first stage that is directed toward the second layer **226** and a second stage that is directed toward the first layer **224**. As shown in FIG. **9**, upon conclusion of the etch back, the junction isolation feature **218J** is exposed in the junction dielectric fin trench **222J**.

(20) Referring now to FIG. 10, with the junction dielectric fin trench 222J exposed, a cladding layer 232 is deposited over the workpiece 200, including over the sidewalls of the junction dielectric fin trench 222J. In some embodiments, the cladding layer 232 may have a composition similar to that of the sacrificial layers 206 or the top sacrificial layer 206T. In one example, the cladding layer 232 may be formed of silicon germanium (SiGe). Their common composition allows selective and simultaneous removal of the sacrificial layers 206 and the cladding layer 232 in a subsequent process. In some embodiments, the cladding layer 232 may be conformally and epitaxially grown using vapor phase epitaxy (VPE) or molecular beam epitaxy (MBE). As shown in FIG. 10, the cladding layer 232 is selectively disposed on exposed sidewall surfaces in the junction dielectric fin trench 222J. Depending on the extent of the selective growth of the cladding layer 232, an etch back process may be performed to expose the junction isolation feature 218J. Still referring to FIG. 10, a third layer 234 and a fourth layer 236 are conformally deposited into the junction dielectric fin trench 222J. The composition and the formation of the third layer 234 may be similar to those of the first layer 224. The fourth layer 236 may include silicon oxide or a silicon oxide containing dielectric material. In some embodiments, the fourth layer 236 may be deposited using CVD, HDPCVD, or flowable CVD (FCVD). In one embodiment, the fourth layer 236 may be deposited using FCVD. After the deposition of the third layer 234 and the fourth layer 236, the workpiece 200 is planarized using a chemical mechanical polishing (CMP) process to expose the top sacrificial layer 206T.

(21) Reference is made to FIG. 11. After the planarization, the first layer 224, the second layer 226, the third layer 234, and the fourth layer 236 are selectively etched back to form recesses and a helmet layer 240 is deposited in such recesses. In some embodiments, the selective etch back may be performed using a dry etch process that may include oxygen, nitrogen, a fluorine-containing gas (e.g., CF_4 , SF_6 , CH_2F_2 , CHF_3 , and/or C_2F_6), a chlorine-containing gas (e.g., Cl_2 , CHCl_3 , CCl_4 , and/or BCl_3), a bromine-containing gas (e.g., HBr and/or CHBr_3), an iodine-containing gas, other suitable gases and/or plasmas, and/or combinations thereof. The helmet layer 240 may include silicon nitride, silicon carbide, silicon carbonitride, silicon oxycarbonitride, aluminum oxide, aluminum nitride, aluminum oxynitride, zirconium oxide, zirconium nitride, zirconium aluminum oxide, hafnium oxide, or a suitable dielectric material. In some embodiments, the helmet layer 240 may be deposited using CVD, HDPCVD, or a suitable deposition technique. After the deposition of the helmet layer 240, the workpiece 200 is planarized using a CMP process to remove excess helmet layer 240 on the cladding layer 232 and the top sacrificial layer 206T. At this juncture, the dielectric fins 230 and the junction dielectric fin 230J are substantially formed. Each of the dielectric fins 230 includes the first layer 224, the second layer 226 over the first layer 224, and the helmet layer 240 over the first layer 224 and the second layer 226. The junction dielectric fin 230J includes the third layer 234, the fourth layer 236, and the helmet layer 240 over the third layer 234 and the fourth layer 236. Each of the dielectric fins 230 is disposed directly over an isolation feature 218. The junction dielectric fin 230J is disposed directly over the junction isolation feature 218J, which is disposed on the deep isolation feature 220, which is disposed over the well junction 2020.

(22) Referring to FIGS. 1 and 12, method 100 includes a block 114 where the top sacrificial layer 206T in the fin-shaped structures 211 are removed. At block 114, the workpiece 200 is etched to selectively remove a portion of the cladding layer 232 and the top sacrificial layer 206T to expose the topmost channel layer 208, without substantially damaging the helmet layer 240 of the dielectric fins 230 and the junction dielectric fin 230J. Because the top sacrificial layer 206T and the cladding layer 232 are formed of silicon germanium (SiGe), the etch process at block 114 may be selective to silicon germanium (SiGe). In some instances, the cladding layer 232 and the top sacrificial layer 206T may be etched using a selective wet etch process that includes ammonium hydroxide (NH_4OH), hydrogen fluoride (HF), hydrogen peroxide (H_2O_2), or a combination thereof. As shown in FIG. 12, after the removal of the top sacrificial layer 206T and

the etching of the cladding layer **232**, the dielectric fins **230** and the junction dielectric fin **230J** rises above the topmost channel layer **208**.

(23) Referring to FIGS. **1**, **13** and **14**, method **100** includes a block **116** where a dummy gate stack **250** is formed over the channel regions of the fin-shaped structures **211**. In some embodiments, a gate replacement process (or gate-last process) is adopted where the dummy gate stack **250** serves as a placeholder for a functional gate structure. Other processes and configuration are possible. As shown in FIG. **13**, the dummy gate stack **250** includes a dummy dielectric layer **242**, a dummy electrode **244** disposed over the dummy dielectric layer **242**. For patterning purposes, a gate top hard mask **248** is deposited over the dummy gate stack **250**. The gate top hard mask **248** may be multi-layer and include a silicon nitride mask layer **246** and a silicon oxide mask layer **247** over the silicon nitride mask layer **246**. The regions of the fin-shaped structures **211** underlying the dummy gate stack **250** may be referred to as channel regions. Each of the channel regions in a fin-shaped structure **211** is sandwiched between two source/drain regions for source/drain formation. In an example process, the dummy dielectric layer **242** is blanketly deposited over the workpiece **200** by CVD. A material layer for the dummy electrode **244** is then blanketly deposited over the dummy dielectric layer **242**. The dummy dielectric layer **242** and the material layer for the dummy electrode **244** are then patterned using photolithography processes to form the dummy gate stack **250**. In some embodiments, the dummy dielectric layer **242** may include silicon oxide and the dummy electrode **244** may include polycrystalline silicon (polysilicon).

(24) Reference is made to FIG. **14**. At block **116**, at least one gate spacer **252** is formed along sidewalls of the dummy gate stacks **250**. The at least one gate spacer **252** may include two or more gate spacer layers. Dielectric materials for the at least one gate spacer **252** may be selected to allow selective removal of the dummy gate stack **250**. Suitable dielectric materials may include silicon nitride, silicon oxycarbonitride, silicon carbonitride, silicon oxide, silicon oxycarbide, silicon carbide, silicon oxynitride, and/or combinations thereof. In an example process, the at least one gate spacer **252** may be conformally deposited over the workpiece **200** using CVD, subatmospheric CVD (SACVD), or ALD.

(25) Referring to FIGS. **1** and **14**, method **100** includes a block **118** where the source/drain regions of the fin-shaped structures **211** are recessed to form source/drain recesses **254**. With the dummy gate stack **250** and the at least one gate spacer **252** serving as an etch mask, the workpiece **200** is anisotropically etched to form the source/drain recesses **254** (or source/drain trenches **254**) over the source/drain regions of the fin-shaped structures **211**. In some embodiments as illustrated in FIG. **14**, operations at block **118** may substantially remove the top portions **212T** of fin-shaped structures **211** in the source/drain regions. In some other alternative embodiments, the source/drain trenches **254** may extend into the base portions **211B**. The anisotropic etch at block **118** may include a dry etch process. For example, the dry etch process may implement hydrogen, a fluorine-containing gas (e.g., CF₄, SF₆, CH₂F₂, CHF₃, and/or C₂F₆), a chlorine-containing gas (e.g., Cl₂, CHCl₃, CCl₄, and/or BCl₃), a bromine-containing gas (e.g., HBr and/or CHBr₃), an iodine-containing gas, other suitable gases and/or plasmas, and/or combinations thereof.

(26) Referring to FIGS. **1**, **15** and **16**, method **100** includes a block **120** where inner spacer features **258** are formed. Referring to FIG. **15**, at block **120**, the sacrificial layers **206** exposed in the source/drain trenches **254** are first selectively and partially recessed to form inner spacer recesses, while the exposed channel layers **208** are substantially unetched. Because the cladding layer **232** and the sacrificial layers **206** share a similar composition, the cladding layer **232** may also be etched at block **120**. In an embodiment where the channel layers **208** consist essentially of silicon (Si), sacrificial layers **206** consist essentially of silicon germanium (SiGe), and the cladding layer **232** consists essentially of silicon germanium (SiGe), the selective and partial recess of the sacrificial layers **206** and the cladding layer **232** may include a SiGe oxidation process followed by a SiGe oxide removal. In that embodiments, the SiGe oxidation process may include use of ozone.

In some other embodiments, the selective recess may include a selective isotropic etching process (e.g., a selective dry etching process or a selective wet etching process), and the extent at which the sacrificial layers **206** and the cladding layer **232** are recessed is controlled by duration of the etching process. The selective dry etching process may include use of one or more fluorine-based etchants, such as fluorine gas or hydrofluorocarbons. The selective wet etching process may include ammonium hydroxide (NH_4OH), hydrogen fluoride (HF), hydrogen peroxide (H_2O_2), or a combination thereof (e.g. an APM etch that includes an ammonia hydroxide-hydrogen peroxide-water mixture). After the formation of the inner spacer recesses, an inner spacer material layer is then conformally deposited using CVD or ALD over the workpiece **200**, including over and into the inner spacer recesses and the space left behind by the removed portion of the cladding layer **232**. The inner spacer material may include silicon nitride, silicon oxycarbonitride, silicon carbonitride, silicon oxide, silicon oxycarbide, silicon carbide, or silicon oxynitride. After the deposition of the inner spacer material layer, the inner spacer material layer is etched back to form inner spacer features **258**, as illustrated in FIG. **16**.

(27) Referring to FIGS. **1** and **17**, method **100** includes a block **122** where n-type source/drain features **260N** and p-type source/drain features **260P** are formed. The n-type source/drain features **260N** and p-type source/drain features **260P** are selectively and epitaxially deposited on the exposed semiconductor surfaces of the channel layers **208** and the substrate **202** in the source/drain trenches **254**. The n-type source/drain features **260N** and p-type source/drain features **260P** may be sequentially deposited using an epitaxial process, such as vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), molecular beam epitaxy (MBE), and/or other suitable processes. In some embodiments, a first masking layer, such as a BARC layer, is deposited to selectively expose the source/drain trenches **254** over the p-well **202P** and then the p-type source/drain features **260P** are deposited over the exposed channel layers **208** and the substrate **202** over the n-well **202N**. After the first masking layer is removed by ashing or stripping, a second masking layer, such as a BARC layer, is deposited to selectively expose the source/drain trenches over the n-well and then the n-type source/drain features **260N** are deposited over the exposed channel layers and the substrate **202** over the p-well **202P**. The second masking layer is then removed. The order of formation of the p-type source/drain features **260P** and the n-type source/drain features may be switched. The n-type source/drain features **260N** may include silicon (Si) doped with an n-type dopant, such as phosphorus (P) or arsenic (As). The p-type source/drain features **260P** may include silicon germanium (SiGe) doped with a p-type dopant, such as boron (B) or gallium (Ga). Doping of the source/drain features may be performed either in situ with their deposition or ex situ using an implantation process, such as a junction implant process.

(28) Referring still to FIGS. **1** and **17**, method **100** includes a block **124** where a contact etch stop layer (CESL) **262** and an interlayer dielectric (ILD) **264** layer are deposited. In some embodiments, the helmet layer **240** of the dielectric fins **230** and the junction dielectric fin **230J** are selectively etched back before the deposition of the CESL **262**. In some instances, the helmet layer **240** may be selectively etched away using buffered hydrofluoric acid (BHF) or diluted hydrofluoric acid (DHF). After the removal of the helmet **240**, the CESL **262** is first conformally deposited over the workpiece **200** and then the ILD layer **264** is blanketly deposited over the CESL **262**. The CESL **262** may include silicon nitride, silicon oxide, silicon oxynitride, and/or other materials known in the art. The CESL **262** may be deposited using ALD, plasma-enhanced chemical vapor deposition (PECVD) process and/or other suitable deposition or oxidation processes. In some embodiments, the ILD layer **264** includes materials such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. The ILD layer **264** may be deposited by spin-on coating, an FCVD process, or other suitable deposition technique. In some embodiments, after formation of the ILD layer **264**, the workpiece **200** may be annealed to improve integrity of the ILD layer **264**. To remove excess

materials and to expose top surfaces of the dummy electrode **244** of the dummy gate stacks **250**, a planarization process (such as chemical mechanical polishing (CMP) process) may be performed to the workpiece **200** to provide a planar top surface. Top surfaces of the dummy electrodes **244** are exposed on the planar top surface.

(29) Referring to FIGS. **1**, **18**, **19**, and **20**, method **100** includes a block **126** where the dummy gate stacks **250** are replaced with a joint gate structure **270**. Operations at block **126** include removal of the dummy gate stacks **250** (shown in FIG. **18**), selective removal of the sacrificial layers **206** in the channel regions to release the channel members **2080** (shown in FIG. **19**), and formation of a joint gate structure **270** to wrap around each of the channel members **2080** (shown in FIG. **20**). Referring to FIG. **18**, the dummy gate stack **250** exposed at the conclusion of block **124** is removed from the workpiece **200** by a selective etch process. The selective etch process may be a selective wet etch process, a selective dry etch process, or a combination thereof. In the depicted embodiments, the selective etch process selectively removes the dummy dielectric layer **242** and the dummy electrode **244** without substantially damaging the helmet layer **240** and the at least one gate spacer **252**. The removal of the dummy gate stack **250** results in a gate trench **266** over the channel regions.

(30) After the removal of the dummy gate stack **250**, channel layers **208**, sacrificial layers **206**, and the cladding layer **232** in the channel region are exposed in the gate trenches **266**. Referring to FIG. **19**, the exposed sacrificial layers **206** between the channel layers **208** and the cladding layer **232** may be selectively removed to release the channel layers **208** as channel members **2080**. In the depicted embodiments where the channel members **2080** resemble a sheet or a nanosheet, the channel member release process may also be referred to as a sheet formation process. Different from channel members of some MBC transistors, the channel members **2080** extend laterally from sidewalls of the dielectric fins **230**. As shown in FIG. **19**, after their release, the channel members **2080** are spaced apart from the junction dielectric fin **230J**. The channel members **2080** are vertically stacked along the Z direction. The selective removal of the sacrificial layers **206** and the cladding layer **232** may be implemented by selective dry etch, selective wet etch, or other selective etch processes. In some embodiments, the selective wet etching includes ammonium hydroxide (NH₄OH), hydrogen fluoride (HF), hydrogen peroxide (H₂O₂), or a combination thereof (e.g. an APM etch that includes an ammonia hydroxide-hydrogen peroxide-water mixture). In some alternative embodiments, the selective removal includes silicon germanium oxidation followed by a silicon germanium oxide removal. For example, the oxidation may be provided by ozone clean and then silicon germanium oxide removed by an etchant such as NH₄OH. With the removal of the sacrificial layers **206** and the cladding layer **232** in the channel region, the dielectric fins **230**, the junction dielectric fin **230J**, the channel members **2080**, the top surface of the base portions **211B**, the inner spacer features **258**, and the junction isolation feature **218J** are exposed in the gate trench **266**.

(31) Referring to FIG. **20**, the joint gate structure **270** is then deposited over the workpiece to wrap around each of the channel members **2080**. The joint gate structure **270** may include an interfacial layer **267** on the channel members **2080** and the substrate **202**, a gate dielectric layer **268** over the interfacial layer **267**, and a gate electrode layer **269** over the gate dielectric layer **268**. In some embodiments, the interfacial layer **267** includes silicon oxide and may be formed as result of a pre-clean process. An example pre-clean process may include use of RCA SC-1 (ammonia, hydrogen peroxide and water) and/or RCA SC-2 (hydrochloric acid, hydrogen peroxide and water). The pre-clean process oxidizes the exposed surfaces of the channel members **2080** and the substrate **202** to form the interfacial layer **267**. The gate dielectric layer **268** is then deposited over the interfacial layer **267** using ALD, CVD, and/or other suitable methods. The gate dielectric layer **268** may include high-k dielectric materials. As used herein, high-k dielectric materials include dielectric materials having a high dielectric constant, for example, greater than that of thermal silicon oxide (~3.9). In one embodiment, the gate dielectric layer **268** may include hafnium oxide. Alternatively,

the gate dielectric layer **268** may include other high-k dielectrics, such as titanium oxide (TiO.sub.2), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta.sub.2O.sub.5), hafnium silicon oxide (HfSiO.sub.4), zirconium oxide (ZrO.sub.2), zirconium silicon oxide (ZrSiO.sub.2), lanthanum oxide (La.sub.2O.sub.3), aluminum oxide (Al.sub.2O.sub.3), zirconium oxide (ZrO), yttrium oxide (Y.sub.2O.sub.3), SrTiO.sub.3 (STO), BaTiO.sub.3 (BTO), BaZrO, hafnium lanthanum oxide (HfLaO), lanthanum silicon oxide (LaSiO), aluminum silicon oxide (AlSiO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), (Ba,Sr)TiO.sub.3 (BST), silicon nitride (SiN), silicon oxynitride (SiON), combinations thereof, or other suitable material. After the formation or deposition of the interfacial layer **267** and the gate dielectric layer **268**, the gate electrode layer **269** is deposited over the gate dielectric layer **268**. The gate electrode layer **269** may be a multi-layer structure that includes at least one work function layer and a metal fill layer. By way of example, the at least one work function layer may include titanium nitride (TiN), titanium aluminum (TiAl), titanium aluminum nitride (TiAlN), tantalum nitride (Ta_N), tantalum aluminum (TaAl), tantalum aluminum nitride (TaAlN), tantalum aluminum carbide (TaAlC), tantalum carbonitride (TaCN), or tantalum carbide (TaC). The metal fill layer may include aluminum (Al), tungsten (W), nickel (Ni), titanium (Ti), ruthenium (Ru), cobalt (Co), platinum (Pt), tantalum silicon nitride (TaSiN), copper (Cu), other refractory metals, or other suitable metal materials or a combination thereof. In various embodiments, the gate electrode layer **269** may be formed by ALD, PVD, CVD, e-beam evaporation, or other suitable process. As shown in FIG. **20**, the joint gate structure **270** wraps around each of the channel members **2080**. It is noted that the joint gate structure **270** does not extend between the dielectric fins **230** and the channel members **2080** as the channel members **2080** are in contact with the dielectric fins **230**.

(32) Referring to FIGS. **1**, **21** and **22**, method **100** includes a block **128** where further processes are performed. Such further processes may include, for example, planarization of the joint gate structure **270**, etching back of the joint gate structure **270**, deposition of a metal cap layer **272**, deposition of a self-aligned cap (SAC) layer **274**, and formation of gate cut features **276**. Referring to FIG. **21**, a planarization process, such as a CMP process, may be performed to the workpiece **200** until the helmet layer **240** is removed and the joint gate structure **270** is divided by the dielectric fins **230** and the junction dielectric fins **230J**. In FIG. **21**, the planarization divides the joint gate structure **270** into a first gate structure **270-1**, a second gate structure **270-2**, a third gate structure **270-3**, and a fourth gate structure **270-4**. The first gate structure **270-1** and the second gate structure **270-2** are separated by a dielectric fin **230**. The second gate structure **270-2** and the third gate structure **270-3** are separated by the junction dielectric fin **230J**. The third gate structure **270-3** and the fourth gate structure **270-4** are separated by a dielectric fin **230**.

(33) To make room for the metal cap layer **272**, the first gate structure **270-1**, the second gate structure **270-2**, the third gate structure **270-3**, and the fourth gate structures **270-4** may be selectively etched back. For example, the selective etch of the gate structures may include a selective wet etch process that uses nitric acid, hydrochloric acid, sulfuric acid, ammonium hydroxide, hydrogen peroxide, or a combination thereof. The metal cap layer **272** is then deposited over the first gate structure **270-1**, the second gate structure **270-2**, the third gate structure **270-3**, and the fourth gate structure **270-4**. In some embodiments, the metal cap layer **272** may include titanium (Ti), titanium nitride (TiN), tantalum nitride (Ta_N), tungsten (W), ruthenium (Ru), cobalt (Co), or nickel (Ni) and may be deposited using PVD, CVD, or metal organic chemical vapor deposition (MOCVD). In one embodiment, the metal cap layer **272** includes tungsten (W) and is deposited by PVD. In some alternative embodiments where the metal cap layer **272** is deposited by MOCVD, the deposition of the metal cap layer **272** may be selectively deposited on the first gate structure **270-1**, the second gate structure **270-2**, the third gate structure **270-3**, and the fourth gate structure **270-4**. After the deposition of the metal cap layer **272**, the SAC layer **274** is deposited over the workpiece **200** by CVD, PECVD, or a suitable deposition process. The SAC layer **274** may include silicon oxide, silicon nitride, silicon carbide, silicon carbonitride, silicon oxynitride,

silicon oxycarbonitride, aluminum oxide, aluminum nitride, aluminum oxynitride, zirconium oxide, zirconium nitride, zirconium aluminum oxide, hafnium oxide, or a suitable dielectric material. Photolithography processes and etch processes are then performed to etch the deposited SAC layer **274** to form gate cut openings to expose the top surfaces of the dielectric fins **230** and the junction dielectric fin **230J**. Thereafter, a dielectric material is deposited and planarized by a CMP process to form the gate cut features **276** in the gate cut openings. The dielectric material for the gate cut features **276** may be deposited using HDPCVD, CVD, ALD, or a suitable deposition technique. In some instances, the gate cut feature **276** may include silicon oxide, silicon nitride, silicon carbide, silicon carbonitride, silicon oxynitride, silicon oxycarbonitride, aluminum oxide, aluminum nitride, aluminum oxynitride, zirconium oxide, zirconium nitride, zirconium aluminum oxide, hafnium oxide, or a suitable dielectric material. In some embodiments, the gate cut feature **276** and the SAC layer **274** may have different compositions to introduce etch selectivity.

(34) An enlarged fragmentary cross-sectional view of the channel region of the semiconductor device **200** in FIG. **21** is shown in FIG. **23**. Each of the first gate structure **270-1**, the second gate structure **270-2**, the third gate structure **270-3**, and the fourth gate structure **270-4** wraps around each of the vertical stacks of channel members **2080**. The first gate structure **270-1** and the second gate structure **270-2** are separated by a dielectric fin **230**. The second gate structure **270-2** and the third gate structure **270-3** are separated by the junction dielectric fin **230J**. The third gate structure **270-3** and the fourth gate structure **270-4** are separated by a dielectric fin **230**. The base portions **211B** under the channel members **2080** are separated by isolation features **218** and the junction isolation feature **218J** as well as lower portions of the dielectric fins **230** and the junction dielectric fins **230J**. The deep isolation feature **220** may have a bulge shape when view along the Y direction. In some embodiments, the deep isolation feature **220** undercuts channel members **2080** wrapped around by the second gate structure **270-2** and the third gate structure **270-3**, as shown by the dotted lines. The deep isolation feature **220** also undercuts the base portions **211B** on both sides of the well junction **2020**. In some instances, the deep isolation feature **220** may have a first height H1 between about 10 nm and about 100 nm along the Z direction and a first width W1 between about 5 nm and about 50 nm along the X direction. The junction isolation feature **218J** may be wider than the isolation features **218**. In some instances, the isolation features **218** may have a second width W2 between about 10 nm and about 20 nm and the junction isolation feature **218J** may have a third width W3 between about 20 nm and about 30 nm. In one embodiment, the first width W1 is greater than the third width W3. In that embodiment, the first width W1 is between about 25 nm and about 50 nm. The isolation features **218** and the junction isolation feature **218J** may have a second height H2 between about 1 nm and about 20 nm. The deep isolation feature **220** and the junction isolation feature **218J** work in synergy to reduce bulk leakage across the well junction **2020**, which is a boundary between the p-well **202P** and the n-well **202N**.

(35) Referring still to FIG. **23**, the first gate structure **270-1** controls a first n-type MBC transistor **302** disposed over the p-well **202P**. The channel members **2080** of the first n-type MBC transistor **302** extend between two n-type source/drain features **260N**. The second gate structure **270-2** controls a second n-type MBC transistor **304** disposed over the p-well **202P**. The channel members **2080** of the second n-type MBC transistor **304** extend between two n-type source/drain features **260N**. The third gate structure **270-3** controls a first p-type MBC transistor **306** disposed over the n-well **202N**. The channel members **2080** of the first p-type MBC transistor **306** extend between two p-type source/drain features **260P**. The fourth gate structure **270-4** controls a second p-type MBC transistor **308** disposed over the n-well **202N**. The channel members **2080** of the second p-type MBC transistor **308** extend between two p-type source/drain features **260P**. When viewed along the Y direction, each of the first gate structure **270-1**, the second gate structure **270-2**, the third gate structure **270-3**, and the fourth gate structure **270-4** includes a fork-like or a fishbone-like structure. For that reason, the first n-type MBC transistor **302**, the second n-type MBC transistor **304**, the first p-type MBC transistor **306**, and the second p-type MBC transistor **308** may be

referred to as fork-sheet transistors or fishbone transistors.

(36) Based on the above discussions, it can be seen that the present disclosure offers advantages. It is understood, however, that other embodiments may offer additional advantages, and not all advantages are necessarily disclosed herein, and that no particular advantage is required for all embodiments. For example, abutting n-well and p-well under fishbone or fork-sheet transistors are separated by a junction isolation feature and a deep isolation feature disposed below the junction isolation feature. The deep isolation feature may have a shape different from that of the junction isolation feature. The deep isolation feature works in synergy with the junction isolation feature to prevent bulk leakage across the well junction.

(37) In one aspect, the present disclosure provides embodiments of a semiconductor structure that includes a substrate having a p-type well or an n-type well, a first base portion over the p-type well, a second base portion over the n-type well, a first plurality of channel members over the first base portion, a second plurality of channel members over the second base portion, an isolation feature disposed between the first base portion and the second base portion, and a deep isolation structure in the substrate disposed below the isolation feature.

(38) In some embodiments, the isolation feature and the deep isolation structure extend through a junction between the p-type well and the n-type well. In some implementations, the deep isolation structure includes silicon oxide, silicon nitride, silicon oxynitride, silicon oxycarbide, silicon carbonitride, or silicon oxycarbonitride. In some instances, a shape of the isolation feature is different from a shape of the deep isolation structure. In some embodiments, the semiconductor structure may further include a first gate structure wrapping around each of the first plurality of channel members, a second gate structure wrapping around each of the second plurality of channel members, and a first dielectric fin disposed between the first gate structure and the second gate structure. In some instances, the first dielectric fin is disposed on the isolation feature. In some embodiments, a first portion of the first gate structure extends between the first dielectric fin and the first gate structure. In some implementations, the semiconductor structure may further include a second dielectric fin in contact with sidewalls of the first plurality of channel members. In some implementations, a second portion of the first gate structure extends between the first dielectric fin and the first base portion. In some instances, the semiconductor structure may further include a second dielectric fin in contact with the first base portion.

(39) In another aspect, the present disclosure provides embodiments of a semiconductor structure that includes a substrate having a p-type well or an n-type well, a first base portion over the p-type well, a second base portion over the n-type well, a first n-type epitaxial feature over the first base portion, a first p-type epitaxial feature over the second base portion, a first isolation feature disposed between the first base portion and the second base portion, and a deep isolation structure in the substrate disposed directly below the first isolation feature.

(40) In some embodiments, the semiconductor structure may further include a first dielectric fin disposed on the first isolation feature. The first dielectric fin is sandwiched between the first n-type epitaxial feature and the first p-type epitaxial feature and a width of the first dielectric fin is smaller than a width of the first isolation feature. In some embodiments, a shape of the first isolation feature is different from a shape of the deep isolation structure. In some instances, the semiconductor structure may further include a third base portion over the p-type well, a second n-type epitaxial feature over the third base portion, and a second isolation feature disposed between the first base portion and the third base portion. A width of the first isolation feature is greater than a width of the second isolation feature. In some instances, the semiconductor structure may further include a second dielectric fin disposed over the second isolation feature. The second dielectric fin is sandwiched between the first n-type epitaxial feature and the second n-type epitaxial feature and a width of the second dielectric fin is substantially identical to a width of the second isolation feature.

(41) In yet another aspect, the present disclosure provides embodiments of a method that includes

receiving a workpiece including a first fin-shaped structure over a p-type well region of a substrate and a second fin-shaped structure over an n-type well region of the substrate, the first fin-shaped structure and the second fin-shaped structure being spaced apart by a junction trench, extending the junction trench further into the substrate to form a deep pocket, forming a deep isolation feature in the deep pocket and an isolation feature in the junction trench, and forming a dielectric fin on the isolation feature such that the dielectric fin is disposed between the first fin-shaped structure and the second fin-shaped structure.

(42) In some embodiments, the method may further include before the extending of the junction trench, forming a liner along sidewalls of the junction trench. In some implementations, the forming of the deep isolation feature and the isolation feature includes depositing a dielectric material into the deep pocket and the junction trench and etching back the liner and the dielectric material to expose sidewalls of the first fin-shaped structure and the second fin-shaped structure. In some instances, the method may further include before the forming of the dielectric fin, depositing a semiconductor cladding layer over the sidewalls of the first fin-shaped structure and the second fin-shaped structure. In some instances, after the forming of the dielectric fin, the dielectric fin is spaced apart from the first fin-shaped structure and the second fin-shaped structure by the semiconductor cladding layer.

(43) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a first base fin, a second base fin, and a third base fin arising from a substrate; a first plurality of channel members over the first base fin; a second plurality of channel members over the second base fin; a first isolation feature disposed between the first base fin and the second base fin; a second isolation feature disposed between the second base fin and the third base fin; and a first dielectric fin disposed over the first isolation feature, wherein the first dielectric fin is sandwiched between the first plurality of channel members and the second plurality of channel members along a direction.
2. The semiconductor structure of claim 1, wherein first isolation feature comprises a first width along the direction, wherein the second isolation feature comprises a second width along the direction, wherein the second width is greater than the first width.
3. The semiconductor structure of claim 2, further comprising: a deep isolation structure disposed below the second isolation feature, wherein the deep isolation structure comprises a third width along the direction, wherein the third width is greater than the second width.
4. The semiconductor structure of claim 1, wherein the substrate comprises a p-type well region and an n-type well region, wherein the first base fin and second base fin are disposed over the p-type well region, wherein the third base fin is disposed over the n-type well region.
5. The semiconductor structure of claim 4, wherein the second isolation feature is disposed between the p-type well region and the n-type well region along the direction.
6. The semiconductor structure of claim 1, further comprises: a first gate structure wrapping around each of the first plurality of channel members; and a second gate structure wrapping around each of the second plurality of channel members.
7. The semiconductor structure of claim 6, wherein the first gate structure is insulated from the

second gate structure by the first dielectric fin.

8. The semiconductor structure of claim 6, further comprising: a second dielectric fin disposed over the second isolation feature, wherein a portion of the second gate structure extends between the second plurality of channel members and the second dielectric fin.

9. The semiconductor structure of claim 8, wherein a portion of the second gate structure extends between the second base fin and the second dielectric fin.

10. The semiconductor structure of claim 8, wherein the portion of the second gate structure lands on a top surface of the second isolation feature.

11. A structure, comprising: a first base fin, a second base fin, and a third base fin arising from a substrate; a first plurality of channel members over the first base fin; a second plurality of channel members over the second base fin; a first isolation feature disposed between the first base fin and the second base fin; a second isolation feature disposed between the second base fin and the third base fin; a first dielectric fin disposed over the first isolation feature; a second dielectric fin disposed over the second isolation feature; a first gate structure wrapping around each of the first plurality of channel members; and a second gate structure wrapping around each of the second plurality of channel members, wherein a portion of the second gate structure extends between the second base fin and the second dielectric fin.

12. The structure of claim 11, wherein the first dielectric fin is sandwiched between the first plurality of channel members and the second plurality of channel members along a direction.

13. The structure of claim 11, wherein the first dielectric fin comprises a first layer and second layer disposed over the first layer, wherein the first layer is in direct contact with the first isolation feature, sidewalls of the first plurality of channel members, sidewalls of the second plurality of channel members, the first gate structure, and the second gate structure, wherein the second layer is spaced apart from the first isolation feature, the sidewalls of the first plurality of channel members, the sidewalls of the second plurality of channel members, the first gate structure, and the second gate structure by the first layer.

14. The structure of claim 13, wherein a dielectric constant of the first layer is greater than a dielectric constant of the second layer.

15. The structure of claim 14, wherein the first layer comprises silicon, silicon nitride, silicon carbide, silicon carbonitride, silicon oxycarbonitride, aluminum oxide, aluminum nitride, aluminum oxynitride, zirconium oxide, zirconium nitride, zirconium aluminum oxide, or hafnium oxide, wherein the second layer comprises silicon oxide, silicon carbide, silicon oxynitride, or silicon oxycarbonitride.

16. The structure of claim 11, wherein top surfaces of the first dielectric fin and the second dielectric fin are higher than topmost ones of the first plurality of channel members.

17. A semiconductor structure, comprising: a substrate comprising a first well region and a second well region; a first base fin over the first well region; a second base fin over the second well region; a first plurality of channel members over the first base fin; a second plurality of channel members over the second base fin; a junction isolation feature disposed between the first base fin and the second base fin along a direction; a dielectric fin disposed over the junction isolation feature; a first gate structure wrapping around each of the first plurality of channel members; and a second gate structure wrapping around each of the second plurality of channel members, wherein the dielectric fin is sandwiched between the first gate structure and the second gate structure along the direction, wherein a portion of the first gate structure and a portion of the second gate structure are in contact with a top surface of the junction isolation feature.

18. The semiconductor structure of claim 17, further comprising: a deep isolation structure disposed below the junction isolation feature.

19. The semiconductor structure of claim 18, wherein the dielectric fin comprises a first width along the direction, wherein the junction isolation feature comprises a second width along the direction, wherein the deep isolation structure comprises a third width along the direction, wherein

the third width is greater than the second width, wherein the second width is greater than the first width.

20. The semiconductor structure of claim 18, wherein the deep isolation structure undercut the first plurality of channel members and the second plurality of channel members.
